Computer Science & IT

Discrete Mathematics

Graph Theory

Lecture No. 04

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Recap of Previous Lecture

- **Topic** Bipartite graph
- **Topic** Complete bipartite graph
- **Topic** Sum of degree theorem
- **Topic** Degree Sequence

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Topics to be Covered

- **Topic** Havel Hakimi's algorithm
- **Topic** Complement of a graph
- **Topic** Graph isomorphism

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Havel Hakimi's Algorithm

Consider the following degree sequence in non-increasing order

Seq 1: $\{S, t_1, t_2, t_3, \ldots, t_S, d_1, d_2, \ldots d_k\}$ {$S, t_1, t_2, \ldots d_1, d_2, \ldots$ etc. are degrees of different vertices}

In order to reduce the size of problem delete the first degree from degree sequence {i.e, delete 'S'} and subtract '1' from the next 'S' degrees in the given sequence and obtain a new degree sequence

i.e. Seq 2: $\{t_1 - 1, t_2 - 1, t_3 - 1, \ldots t_S - 1, d_1, d_2, \ldots d_k\}$

"Seq 1" can represent a simple non-directed graph if and only if "Seq 2" can represent a simple non-directed graph

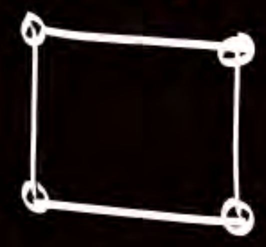
If we are not able to visualize the graph w.r.t. "Seq 2" as well, then rearrange the degrees of "Seq 2" in non-increasing order and repeat the process

Keep repeating until you are not able to visualize the graph

Q:- Check whether the following degree sequence can represent a simple non-directed graph or not?

Seq 1:- $\{ \cancel{3}, \underbrace{3, 3, 3}, 2 \}$

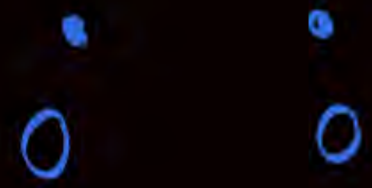
Seq 2: $\{ \cancel{2}, \underbrace{2, 2}, 2 \}$ =



Seq 3: $\{ 1, 1, 2 \}$ =

Rearranged Seq 3: $\{ \cancel{2}, \underbrace{1, 1} \}$

Seq 4: $\{ 0, 0 \}$ =



Final sequence can represent a simple non-directed graph, ∴ Original sequence will also represent a simple non-directed graph

#Q. Which of the following degree sequences represent a simple non directed graph? { i.e. Which of the following is/are graphic }

$S1 = \{6, 6, 6, 6, 4, 3, 3, 0\}$

$S2 = \{6, 5, 5, 4, 3, 3, 2, 2, 2\}$

A. Only S1

B. Only S2 ✓

C. Both S1 and S2

D. Neither S1 nor S2

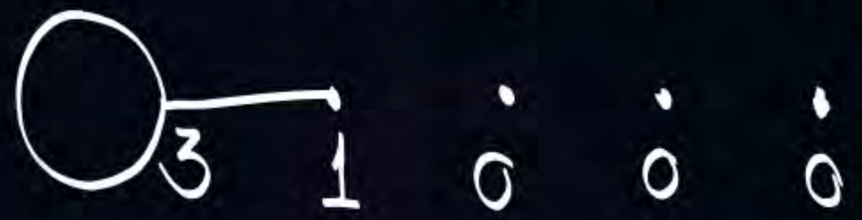
Seq 1 $S1 = \{\cancel{6}, \underbrace{6, 6, 6, 4, 3, 3}, 0\}$

Seq 2 $= \{\cancel{5}, \underbrace{5, 5, 3, 2, 2}, 0\}$

Seq 3 $= \{\cancel{4}, \underbrace{4, 2, 1, 1}, 0\}$

Seq 4 $= \{3, 1, 0, 0, 0\} \Rightarrow$ Final deg sequence does not represent a simple non-directed graph

$\therefore$ Original seq. also will not represent any simple non-directed graph



"Seq 1" $S2 = \{\cancel{6}, \underbrace{5, 5, 4, 3, 3, 2}, 2, 2\}$

Seq 2 $= \{4, 4, 3, 2, 2, 1, 2, 2\}$

Rearranged Seq 2 $= \{\cancel{4}, \underbrace{4, 3, 2, 2}, 2, 2, 1\}$

Seq 3 $= \{3, 2, 1, 1, 2, 2, 1\}$

Rearranged Seq 3 $= \{\cancel{3}, \underbrace{2, 2, 2}, 1, 1, 1\}$

Seq 4 $= \{1, 1, 1, 1, 1, 1\}$

"Seq 4" can represent a simple non-directed graph, ∴ original seq. can also represent a simple non-directed graph.

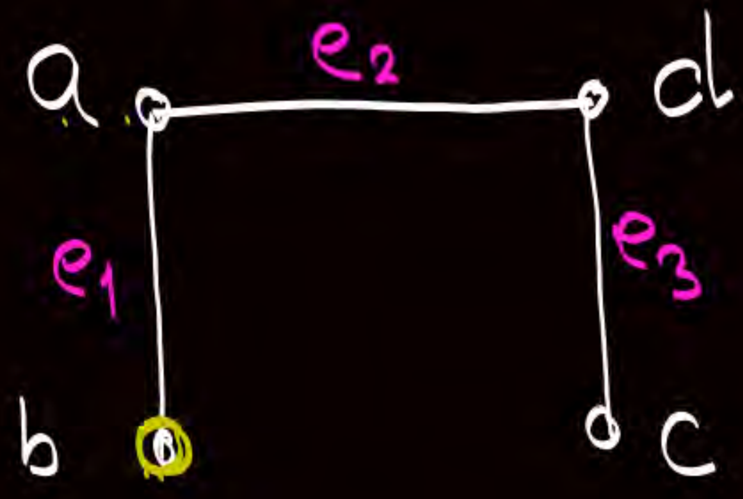
Topic : Complement of a graph

{ Complement can be defined only for a simple graph }

Let G be a simple graph with n-vertices, then complement of graph G is a simple graph with same n-vertices as of G but an edge is present in complement of graph G if and only if that edge is not present in G.

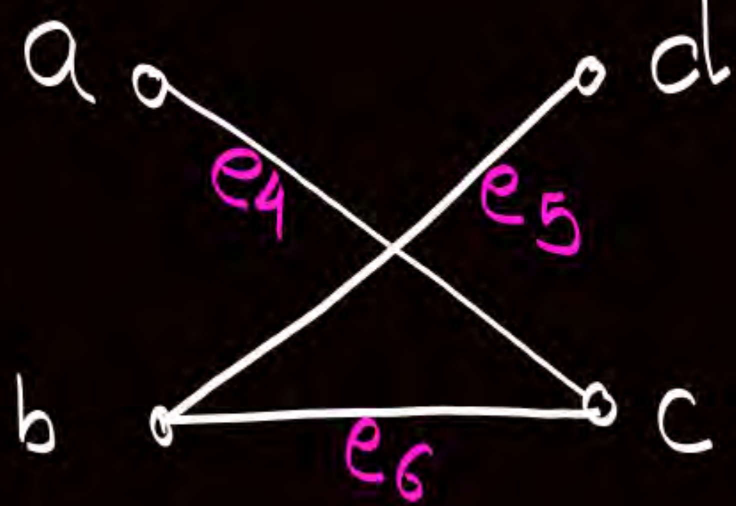
Complement of graph G is denoted by $\overline{G}$

eg:



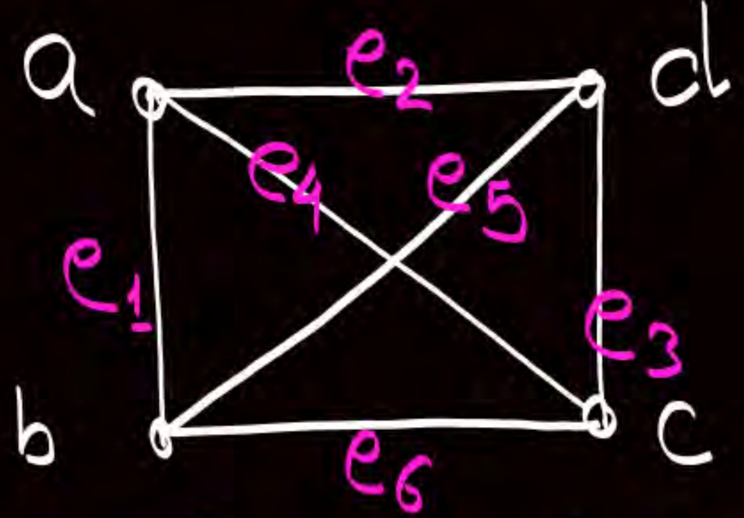
G

$\cup$



$\overline{G}$

$\equiv$



$G \cup \overline{G}$ is a complete graph

$G = (V, E_1)$

$V = \{a, b, c, d\}$

$E_1 = \{e_1, e_2, e_3\}$

$\overline{G} = (V, E_2)$

$V = \{a, b, c, d\}$

$E_2 = \{e_4, e_5, e_6\}$

$G \cup \overline{G} = (V \cup V, E_1 \cup E_2)$

$V \cup V = \{a, b, c, d\}$

$E_1 \cup E_2 = \{e_1, e_2, e_3, e_4, e_5, e_6\}$

Note

* Let G be a simple graph with n- vertices and $\overline{G}$ is complement of graph G.

then,

$$\boxed{G \cup \overline{G} = K_n}$$

$$\boxed{|V(G)| = |V(\overline{G})|}$$

$$\boxed{E(G) \cap E(\overline{G}) = \phi}$$

$$\therefore |E(G \cup \overline{G})| = |E(G) \cup E(\overline{G})|$$

$$\Downarrow$$

$$|E(K_n)| = |E(G)| + |E(\overline{G})|$$

$$|E(G)| + |E(\overline{G})| = |E(K_n)|$$

$$|E(G)| + |E(\overline{G})| = \frac{n(n-1)}{2}$$

Q: Let G be a simple graph with n-vertices and '21' edges. If there are '24' edge in Complement of graph G, then find the value of 'n'.

$$\left.\begin{array}{l}|E(G)| = 21 \\ |E(\bar{G})| = 24\end{array}\right\} \Rightarrow |E(G)| + |E(\bar{G})| = \frac{n(n-1)}{2}$$

$$21 + 24 = \frac{n(n-1)}{2}$$

$$45 = \frac{n(n-1)}{2}$$

$$n(n-1) = 90$$

$$\boxed{n = 10} \quad \underline{\text{Ans}}$$

Note:-

* Let G be a simple graph with 'n' vertices and following degree sequence.

degree seq. of $G = \{x_1, x_2, x_3, \ldots, x_n\}$

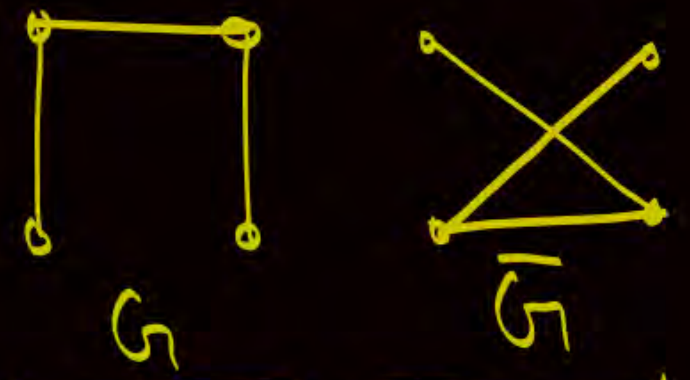
then,

degree sequence of $\overline{G} = \{(n-1-x_1), (n-1-x_2), (n-1-x_3), \ldots, (n-1-x_n)\}$

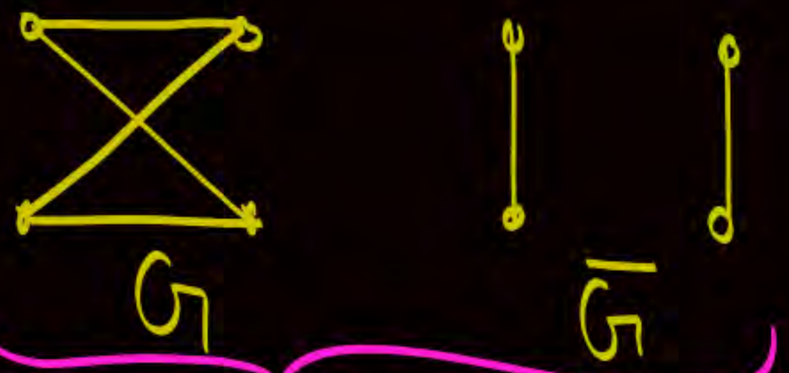
Note:-

Let G is a simple graph and $\bar{G}$ is Complement of graph G, then at least one of G or $\bar{G}$ is a Connected graph

↳ i.e, both of them can never be disconnected



both connected



One is connected and another is disconnected

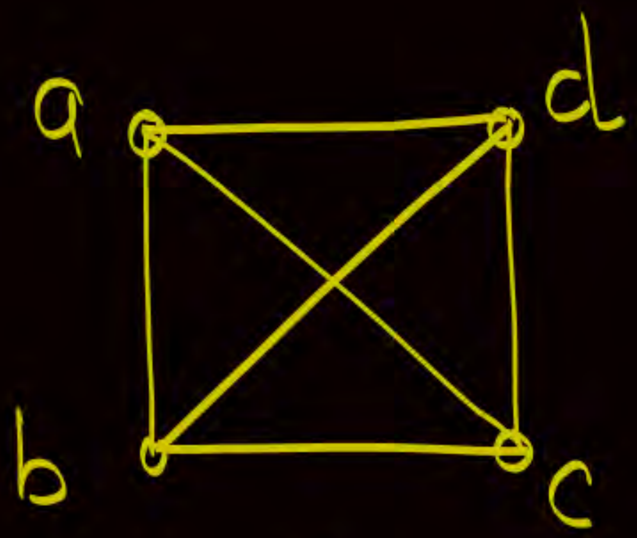
Topic : Graph Isomorphism

Two graphs are said to be isomorphic if their properties are identical (name and look need not be same)

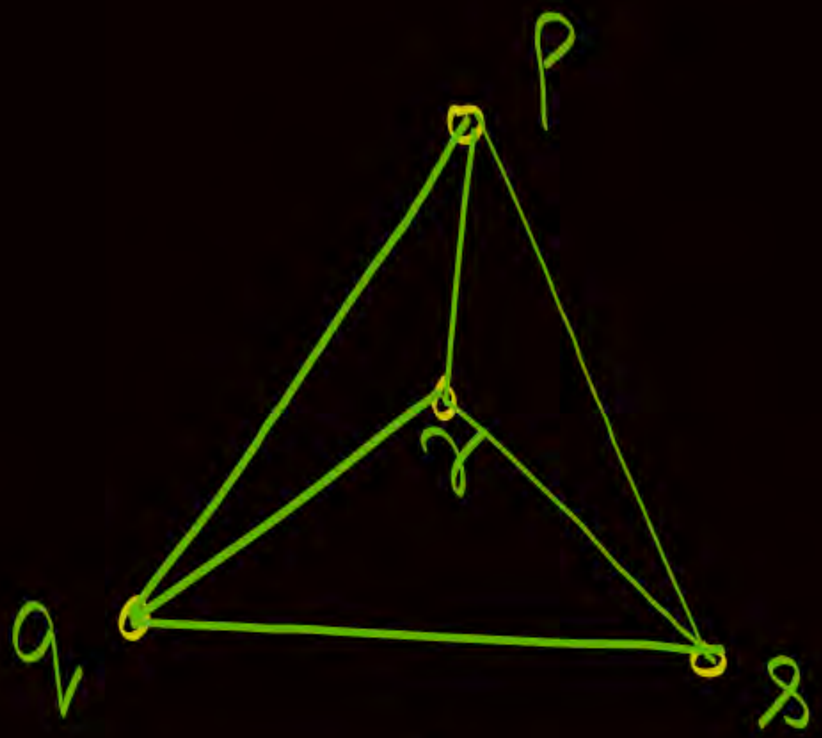
Two graphs G and G' are said to be isomorphic if there exists a function f: V(G)→V(G') such that

I. f is bijective i.e, $|V(G)| = |V(G')|$ and for every vertex of set V(G) there is a unique image in V(G')
II. f preserves adjacency of vertices

i.e. two vertices of graph G are adjacent if and only if their corresponding images in G' are adjacent $\left\{ \therefore |E(G)| = |E(G')| \right.$



G_1



G_2

$V(G_1) = \{a, b, c, d\}$

$V(G_2) = (p, q, r, s)$

$$\left.\begin{aligned} f(a) &= p \\ f(b) &= q \\ f(c) &= r \\ f(d) &= s \end{aligned}\right\}$$

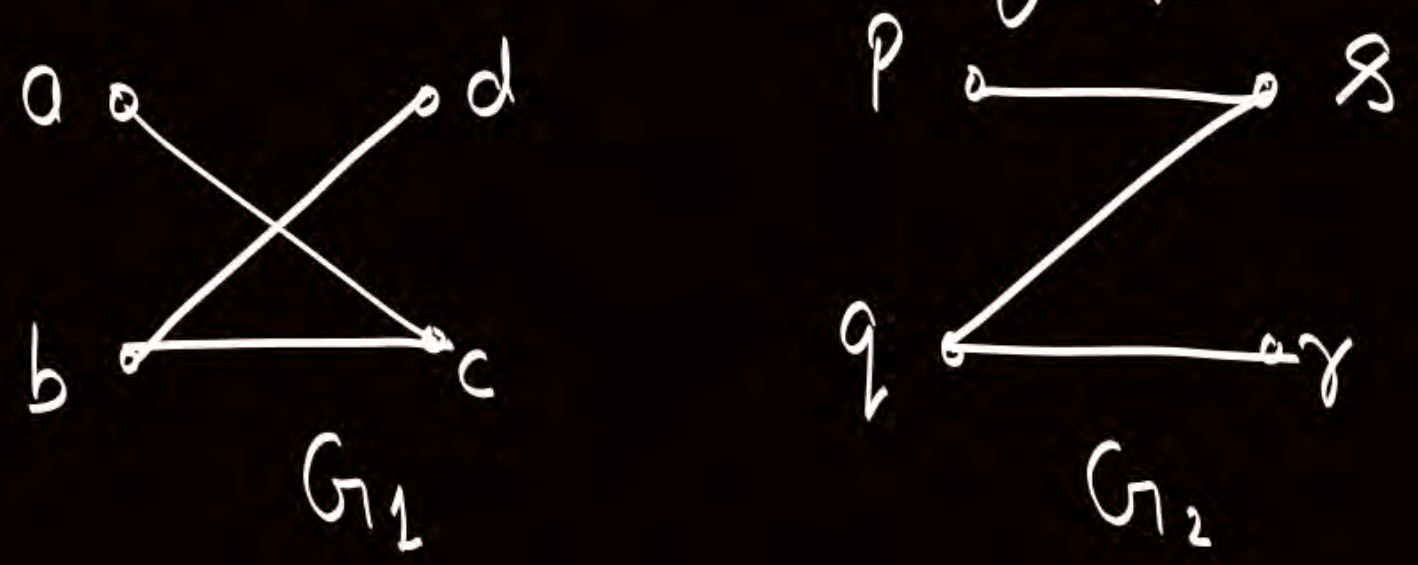
This bijective function exist an, it preserves the adjacency of vertices

$\therefore$ G_1 and G_2 are isomorphic

G is isomorphic to G' is denoted by

$$G \cong G'$$

Q: Consider two graphs G_1 and G_2.
Check whether the graphs are isomorphic or not

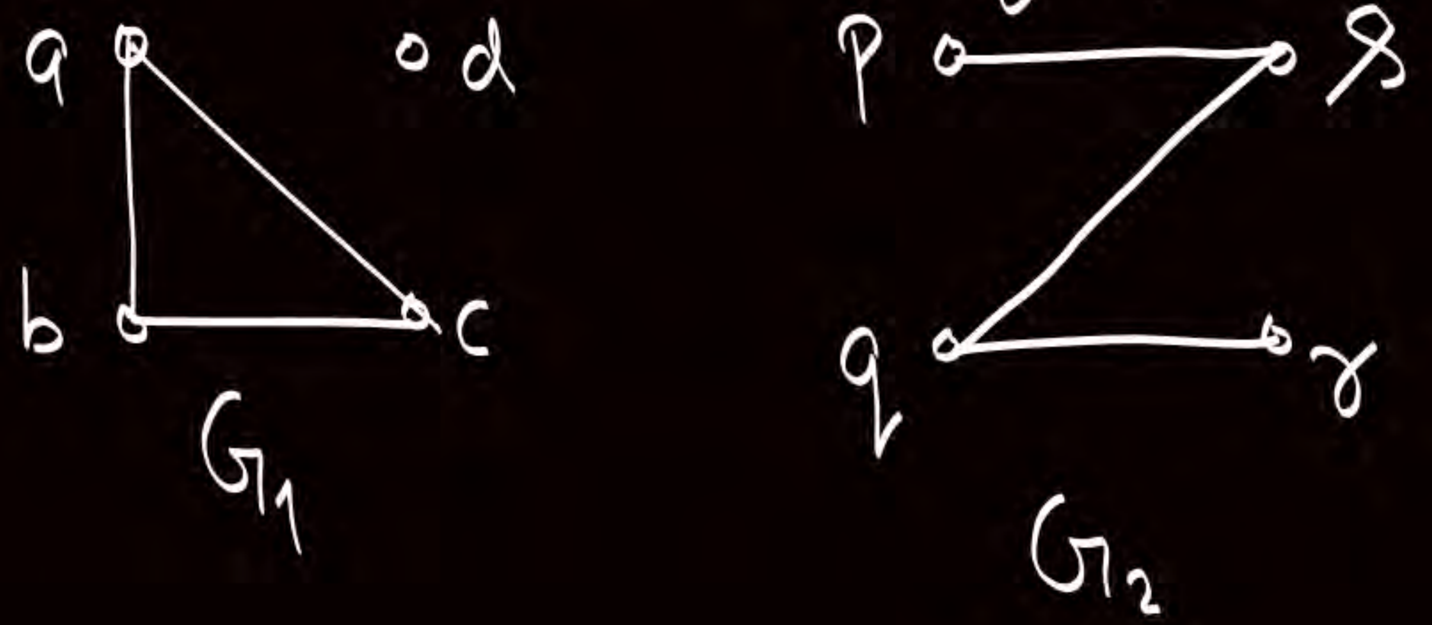


$|V(G_1)| = |V(G_2)|$
and $|E(G_1)| = |E(G_2)|$ } $\therefore$ Graphs may be isomorphic (not necessarily)

$f(a) = p$
$f(b) = q$
$f(c) = s$
$f(d) = r$
} This bijection is possible, s.t. it preserves the adjacency of vertices

$\therefore$ G_1 & G_2 are isomorphic

Q: Consider two graphs G_1 and G_2.
Check whether the graphs are isomorphic or not



$$\left.\begin{array}{l}|V(G_1)| = |V(G_2)| \\ \text{and } |E(G_1)| = |E(G_2)|\end{array}\right\}$$

Condition holds true, but graphs are not isomorphic

Topic : Graph Isomorphism

If $G \cong G'$, then following conditions must hold true.

These four conditions are necessary cond^n for two graphs to be isomorphic

If any of the four cond^n is dis-satisfied then graphs cannot be isomorphic

1. $|V(G)| = |V(G')|$
2. $|E(G)| = |E(G')|$
3. Degree sequence of G = Degree sequence of G'
4. If some 'k' vertices $v_1, v_2, \dots v_k$ in graph G form a cycle of length 'k', then their corresponding images in G' i.e. $f(v_1), f(v_2), f(v_3), \dots f(v_k)$ must also form a cycle of length 'k'.

H.W.

Q: Consider two graphs G_1 & G_2

G_1 G_2

Check whether the graphs are isomorphic or not?

2 mins Summary

Topic Havel Hakimi's algorithm

Topic Complement of a graph

Topic Graph isomorphism

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THANK - YOU

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